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Enhancing energy recovery from corn straw via two-stage anaerobic digestion with stepwise microaerobic hydrogen fermentation and methanogenesis

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ABSTRACT

Anaerobic digestion (AD) technology attracts increasing concerns for organic waste disposal, renewable energy production as well as organic fertilizer preparation. However, the AD of cellulosic substrate is limited by low efficiency. Microaeration has recently been used to improve the AD efficiency. However, the inhibition of methanogenesis by microaeration limited its application. To enhance the energy recovery from AD of corn straw and avoid the inhibition of microaeration on methanogenesis, two-stage AD including microaerobic hydrogen fermentation (stage I) and methanogenesis (stage II) was investigated. Results showed microaeration had beneficial effects on hydrogen fermentation, methane production and energy recovery depending on the oxygen loading level. At the optimized initial pH of 7.5 for hydrogen production, microaerobic hydrogen fermentation achieved the maximum hydrogen yield of 24.3 mL/g VS (volatile solid, 37 °C and normal pressure) with the oxygen loading of 5 mL/g VS, which was 70% higher than that without microaeration. Overall, the two-stage AD performed better in methane yield and energy recovery than one-stage AD. When the oxygen loading during hydrogen fermentation was 10 mL/g VS of corn straw, the maximum total energy recovery of 10.6 kJ/g VS was obtained, which was 18% higher than that of one-stage AD. The two-stage AD consisting of microaerobic hydrogen fermentation (stage I) and methanogenesis (stage II) is a promising way to enhance the energy recovery from AD of cellulosic substrates like corn straw.

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1. Introduction

Anaerobic digestion (AD) technology has been widely applied in the disposal of organic waste (Ye et al., 2018). Given its efficient performance in waste reduction and energy recovery, AD receives increasing interests in the research community and the public worldwide (Nielfa et al., 2015). The high methane content (around 50–70%) in biogas makes it suitable for replacing traditional energy sources (Esen and Yuksel, 2013; Baena-Moreno et al., 2019). Agricultural wastes, such as corn and wheat straw are abundant and rich in carbohydrate contents, which have been demonstrated as potential sources for bioenergy production via AD technology (He et al., 2018; Mahmoodi et al., 2018). However, the compact structure and complex composition of cellulosic agriculture wastes lead to low AD efficiency, which is one of the challenges for the AD of agricultural wastes. Measures including substrate pretreatment (Wang et al., 2017), co-digestion (Jin et al., 2018; Thanarasu et al., 2018), bioaugmentation (Zhang et al., 2015) and combination of these technologies (Mulat et al., 2018) have been extensively taken to improve the AD performance. However, some inherent shortcomings such as extra energy input, chemicals requirement, and/or the suitability limitation for scale-up exist in these measures (Lim and Wang, 2013). For instance, the thermal/chemical pretreatments need large amount of energy/chemical input, which can also generate inhibitory and/or biologically non-degradable

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compounds; co-digestion is not efficient enough in the improvement of methane yield, which is also limited by the substrate supply; the long-term and steady subculture of selected microorganism is a problem for bioaugmentation technology (Wei, 2016). Therefore, explorations for simple and practical methods to improve the efficiency are urgently needed to further advance the applications of AD technology.

Recently, microaeration, which means introducing of limited oxygen/air into the anaerobic digester (ranged from 0.005 to 5 L O_2/L reactor/day according to different studies) (Nguyen and Khanal, 2018), has been found to be an efficient method to improve the AD performance of organic wastes (Fu et al., 2016b; Tsapekos et al., 2017). As reported, microaeration could regulate the microbial community structure by stimulating the growth of certain facultative microorganism, thus leading to higher hydrolysis efficiency (Fu et al., 2016b; Lim et al., 2014). Microorganisms affiliating to phylum *Firmicutes* were reported to play a key role in microaeration (Fu et al., 2016a; Lim et al., 2014). However, the methanogenesis of AD, which is carried out by methanogens, is sensitive to oxygen. Though some researchers reported methanogenic microorganisms such as *Methanobacterium* and *Methanosarcina* can tolerate to limited oxygen (dissolved oxygen up to 5.6 mg/L) (Nguyen and Khanal, 2018; Botheju and Bakke, 2011), microaeration during AD could inhibit the methanogenesis process at certain level of oxygen loading. It is challenging to balance the positive effect and negative effect of microaeration during AD. As such, two-stage AD, in which the AD process is separated into the hydrolysis/acidogensis (stage I) and methanogenesis (stage II) that are optimized individually (Algapani et al., 2018; Tijani et al., 2018) could be a potential solution for the problem of the application of microaeration in AD. According to a two-stage AD including microaerobic hydrogen fermentation and methanogenesis, the inhibition of oxygen on methanogenesis could be avoid by preventing the contiguity of oxygen and methanogens. Improved hydrogen yield under microaeration condition has also been reported in previous study (Li et al., 2016). Actually, the hydrogen fermentation stage of two-stage AD could also be considered as biological pretreatment method for organic solid waste, during the hydrogen fermentation hydrogen energy recovery as well as substrate pretreatment could be achieved simultaneously (Ariunbaatar et al., 2014). Microaeration during hydrogen fermentation stage could obtain a higher hydrolysis of complex substrates due to the higher production of hydrolytic enzymes, thus benefit the following methane fermentation (Lim and Wang, 2013). Through two-stage AD might operate a little difficultly, two-stage AD of cellulosic substrate was reported to be more stable and efficient than one-stage AD, which recovered more energy from substrates (Nathao et al., 2013).

In order to improve the stability and energy recovery of the AD process while avoiding the inhibition of oxygen on methanogenesis, two-stage AD of corn straw including microaerobic hydrogen fermentation (stage I) and methanogenesis (stage II) was examined in this study. The initial pH and the oxygen loading for microaerobic hydrogen production were optimized. The traditional two-stage and one-stage AD without microaeration were also performed as controls to reflect the effects of microaeration and two-stage approaches on the AD process. In addition, energy recovery from AD of corn straw was also calculated based on the hydrogen and methane yields. The implementation of this study could offer an efficient and simple way to enhance the energy recovery from cellulosic substrates like corn straw through AD technology.

2. Materials and methods

2.1. Substrate and inoculum

Air-dried corn straw (dried in the field) was collected from the field of Lianyungang (Jiangsu province, China) as substrate. Before used, the corn straw was chopped and sieved to size of less than 0.5 cm by 0.5 cm sieve. Anaerobic sludge collected from an anaerobic digester of Tuandao Water Treatment Plant that mainly treats urban sewage (Qingdao, Shandong province, China, mainly treats urban sewage) was used as inoculum. The collected inoculum was concentrated using centrifuge method and stored in a refrigerator at 4 °C. Before fermentation experiment, the inoculum was activated in a shaking water bath at 37 °C with the shaking speed of 120 rpm for 3 days. The characteristics of the substrate and inoculum are shown in Table 1.

2.2. Experimental designs

2.2.1. Optimization of initial pH for hydrogen production

Optimization of initial pH for hydrogen production was conducted in 300 mL glass bottles with a working volume of 150 mL. Prior to the start of the hydrogen fermentation of two-stage AD, the inoculum was firstly heated at 80 °C for 15 min to suppress the activity of methanogens and enrich the hydrogen-producing microorganisms (Rafieenia et al., 2017). During the optimization of initial pH for hydrogen production, 5.33 g (TS) corn straw and 1.33 g (TS) heat-treated anaerobic sludge were mixed in the sterile bottles, and the working volumes were adjusted to 150 mL by adding distilled water. The designed initial pH for optimization included 4.5, 5.5, 6.5, 7.5, and 8.5. The pHs of the fermentation systems were adjusted to designed initial pH with NaOH (2 M) and HCl (2 M) solutions, and the pH values were no longer adjusted during the hydrogen fermentation process. All the bottles were flushed with N_2 for 5 min to remove the oxygen and sealed with rubber stoppers. Then, all the bottles were incubated in a shaking water bath at 37 °C with the shaking speed of 120 rpm until the daily biogas yield was less than 1% of the cumulative biogas yield.

Table 1
Characteristics of the substrate and inoculum (tested in triplicate).

Characteristics	Substrate (Corn straw)	Inoculum (Anaerobic sludge)
Moisture content (%)	11.97 ± 0.18	79.90 ± 1.00
Total solid (TS, %)	88.03 ± 0.18	20.10 ± 1.00
Volatile solid (VS, % of TS)	87.43 ± 1.5	50.90 ± 1.91
C (% of TS)	36.66 ± 1.24	–
N (% of TS)	1.27 ± 0.14	–
H (% of TS)	4.87 ± 0.42	–
C/N	28.87 ± 1.91	–
Cellulose (% of TS)	41.87 ± 0.57	–
Hemicellulose (% of TS)	22.02 ± 0.71	–
Lignin (% of TS)	10.88 ± 0.16	–

Note: "–" means this parameter was not tested.

2.2.2. Microaeration during the hydrogen fermentation of two-stage AD

With the predetermined optimal initial pH in section 2.2.1, the effects of microaeration on the hydrogen fermentation of the two-stage AD were studied. The procedure of this process was in accordance with Section 2.2.1. The designated total amount of oxygen during the whole microaerobic hydrogen fermentation process were 0, 5, 10, 20, and 40 mL/g VS (total VS of corn straw and inoculum, marked as Two-stage 0, Two-stage 5, Two-stage 10, Two-stage 20, and Two-stage 40, respectively). The Two-stage 0 meant two-stage AD without microaeration, which was conducted as a control to reflect the effect of microaeration. The needed oxygen was calculated based on the designed oxygen load and the mass of corn straw added, then injected into the sealed reactors daily using syringe. The fermentation time was 5 days according to the results of section 2.2.1.

2.2.3. Methane fermentation of one- and two-stage AD

After the hydrogen fermentation stage, the bottles were opened and re-inoculated with 1.33 g fresh anaerobic sludge (TS) to initiate the methane fermentation stage. The working volumes were adjusted to 200 mL by adding distilled water and the pHs were adjusted to 7.5 with NaOH (2 M) and HCl (2 M). Then, all the bottles were flushed with N_2 for 5 min to remove the oxygen and sealed with rubber stoppers. The one-stage AD of corn straw was also run in parallel, where 5.33 g corn straw (TS) and 2.66 g anaerobic sludge (TS) were charged into the bottles to which a certain amount of distilled water was added to increase the working volume to 200 mL. The pH adjustment and bottle purging were done by following the same procedures used for the two-stage AD. Finally, all the bottles were incubated at 37 °C with 120 rpm in a shaking water bath for methane fermentation until the daily biogas yield was less than 1% of the cumulative biogas yield. Each treatment was run in triplicate unless specified, otherwise.

2.3. Data analysis

The daily biogas yield in volume was measured by water replacement method. The methane/hydrogen yield during the AD process was calculated as Eq. (1).

$$V_t = \left[C_t \times V_{biogas} + (C_t - C_{t-1}) \times V_{headspace}\right] \Big/ VS_{substrate} \quad (1)$$

Where:

t= time (day)
V_t = the methane and/or hydrogen yield (mL/g VS) of day t
C_t = the methane and/or hydrogen concentration (%) in biogas of day t
C_{t-1} = the methane and/or hydrogen concentration (%) in biogas of day t-1
V_{biogas} = the biogas yield (mL) of day t
$V_{headspace}$ = bottle's headspace volumes (mL), which are 150 and 100 mL during the hydrogen and methane fermentation, respectively.
$VS_{substrate}$ = the VS of corn straw added (g)

The VS degradation rate during one/two stage AD of corn straw was calculated as Eq. (2).

$$\text{VS degradation rate} = (M_{added} - M_{remained})/\, M_{added} \quad (2)$$

Where:

M_{added} = Mass of corn straw added (g, based on VS)
$M_{remained}$ = Mass of corn straw remained after AD (g, based on VS)

The energy recovery from corn straw was calculated as Eq. (3).

$$E_r = 12.74 \times V_{hydrogen} + 35.16 \times V_{methane} \quad (3)$$

Where:

E_r = the energy recovery from corn straw (kJ/g VS)
$V_{hydrogen}$ = the hydrogen production from corn straw (L/g VS)
$V_{methane}$ = the methane production from corn straw (L/g VS)

The lower heating value of H_2 (12.74 kJ/L) and CH_4 (35.16 kJ/L) were used to calculate the E_r (Gioannis et al., 2017).

The methane characteristics were described by fitting the data using *the Modified Gomperz Equation* Eq. (3) with origin 8.0 (OriginLab, USA) (Wang et al., 2018; Zhang et al., 2014).

$$P_t = P\exp\left\{-\exp\left[\frac{R_m * e}{P} * (\lambda - t) + 1\right]\right\} \quad (4)$$

Where:

P_t = the cumulative methane yield (mL/g VS) of day t
P = the total methane production potential (mL/g VS)
R_m = the maximum methane production rate (mL/g VS per day)
e = 2.718
λ = the lag-phase time (day)
t= time (day)

2.4. Analytical methods

The methane and hydrogen concentrations in biogas were measured with a gas chromatograph (GC, SP6890, Shandong Lunan Inc., China) equipped with a thermal conductivity detector (TCD) and Porapak Q chromatographic column (180 cm × Ø3mm). The carrier gases for methane and hydrogen tests were argon and nitrogen, respectively. The temperatures of the injector, detector, and oven were 100, 100 and 80 °C, respectively. The TS and VS were measured according to a standard method (APHA, 2006). The cellulose, hemicellulose and lignin contents of corn straw were measured by using the method of Goering and van Soest (1970). Element and pH were measured with elemental analyzer (Vario EL III elemental analyzer) and pH meter (B-212, HORIBA), respectively. To measure volatile fatty acids (VFAs), the liquid samples were firstly centrifuged at 10,000 rpm for 5 min, and the supernatants were filtered through a 0.45 μm glass microfiber filter and acidized to pH of 2.5 using 2 M HCl before GC analysis. VFAs were analyzed on a GC (GC-2014, SHIMADZU) equipped with Innowax column (30m*0.25 mm*0.25m) and a flame ion-ization detectors (FID). The temperatures of the injector and detector were 220 °C, respectively. The temperature of the oven was set to increase from 25 °C to 180 °C at the speed of 10 °C/min.

3. Results and discussion

3.1. Anaerobic hydrogen yield at different initial pH

The pH value has a great influence on the growth of acidogenic microorganisms and play a crucial role in the acidogenesis step of two-stage AD system (Ravi et al., 2018). In this study, the initial pH was optimized at the initial pH values of 4.5, 5.5, 6.5, 7.5, and 8.5 toward the highest hydrogen yield. The hydrogen fermentation was

conducted until the daily biogas yield was less than 1% of the cumulative biogas yield. Actually, after 5th day of fermentation, all of the hydrogen yields decreased rapidly (there were no biogas yield at the initial pH of 4.5, 5.5 and 6.5). Therefore, the running time for hydrogen fermentation in this study was set as 5 days. The temporal cumulative hydrogen yield with different initial pH values are shown in Fig. 1. The initial pH showed significant effect on the hydrogen yield of corn straw. The 5-day cumulative hydrogen yields at the initial pH values of 4.5, 5.5, 6.5, 7.5, and 8.5 were 6, 5.9, 9.4, 13.7, and 6.5 mL/g VS (37 °C and normal pressure) corn straw, respectively. Statistical analysis by T-test: a pairwise comparisons was conducted with the software of Microsoft Office Excel 2016). Results showed the hydrogen yields at the initial pH values of 4.5, 5.5, and 8.5 had no significant difference with each other ($p > 0.05$). The hydrogen yields at the initial pH of 6.5 and 7.5 were statistical higher than others. When the initial pH increased from 7.5 to 8.5, the hydrogen yield decreased significantly ($p < 0.5$). The lower Hydrogenase activity at higher pH value could be the reason for this phenomenon (Kim et al., 2011).

The results obtained in this study are similar to the previous study conducted by Kim et al. (2011), who reported the optimized initial pH for hydrogen fermentation of food waste was 8.0. However, due to the complex structure corn straw, the maximum hydrogen yield from corn straw at the initial pH of 7.5 was lower than that from food waste (Nathao et al., 2013) or waste activated sludge (Wan et al., 2016). In this research, pH 7.5 was selected for all the following studies.

3.2. Effect of microaeration on hydrogen fermentation of two-stage AD

3.2.1. Hydrogen production

Microaerobic hydrogen fermentations were conducted at the initial pH value of 7.5 that optimized from the Section 3.1. Microaerobic condition is defined by Nguyen and Khanal (2018) as introduction of small amount of air or oxygen (typically ranged from 0.005 to 5 L O_2/L reactor/day) into an anaerobic system. In this study, microaerobic hydrogen fermentation lasted for 5 days, during which the microaerobic condition was created by daily oxygen injection with a syringe into the headspace of the bottles. The cumulative hydrogen yields at different oxygen load were shown in Fig. 2. Statistical analysis of the effects of oxygen loads on hydrogen yields were also analyzed by T-test: a pairwise comparisons. Results showed the oxygen injected into the system affected the hydrogen yield significantly ($p < 0.05$). The cumulative hydrogen yields ranked in the order of Two-stage 5 > Two-stage 10 > Two-stage 0 > Two-stage 20 > Two-stage 40. Maximum hydrogen yield of 22.7 mL/g VS (37 °C and normal pressure) was obtained with the oxygen loading of 5 mL/g VS, which is 67% higher than that of Two-stage 0 (anaerobic hydrogen fermentation without microaeration). Improved hydrogen yield under microaerobic condition was also reported by Li et al. (2016) who indicated that the microaerobic condition could benefit the hydrolysis of the substrate, leading to higher hydrogen yield. Improved hydrolysis process under microaerobic condition was widely reported (Montalvo et al., 2016). The microaerobic condition could stimulate the growth of anaerobic/facultative microorganisms to produce more extracellular enzymes, leading to higher hydrolysis efficiency (Girotto et al., 2016; Lim et al., 2014). However, excessive oxygen loadings over 5 mL/g VS had insignificant positive (i.e., 10 mL/g VS on day 5, compared to Two-stage 0) and even inhibitory effect (e.g., 20 and 40 mL/g VS) on hydrogen yield. As shown in Fig. 2, the hydrogen yield at the oxygen loading of 10 mL/g VS had no obvious difference with that of Two-stage 0 ($p > 0.05$). Furthermore, hydrogen yields at the oxygen loadings of 20 and 40 mL/g VS were significantly lower than that of Two-stage 0 ($p < 0.05$). Especially at the oxygen loading of 40 mL/g VS, the hydrogen yield was extremely inhibited and was only 2.8 mL/g VS (37 °C and normal pressure), which means the activities of hydrogenogens were significantly inhibited. Hydrogenogens are responsible for the conversion of simple sugars to hydrogen and organic acid during the hydrogen fermentation stage of AD process (Lin et al., 2018), but most of them are sensitive to oxygen. Therefore, the hydrogen production process could be inhibited in the presence of overdosed oxygen (e.g., oxygen

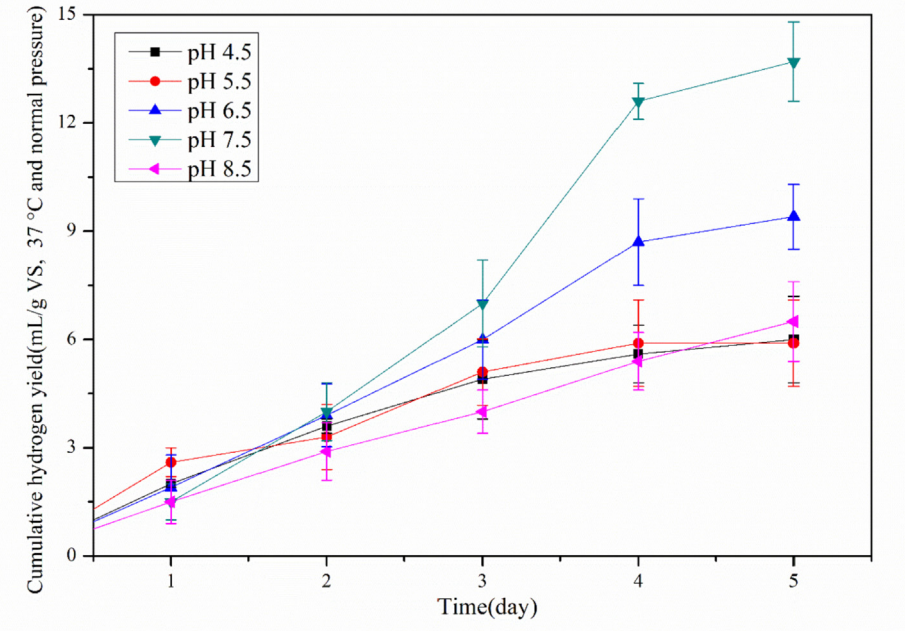


Fig. 1. Cumulative hydrogen yield at different initial pHs during the hydrogen fermentation stage of two-stage AD.

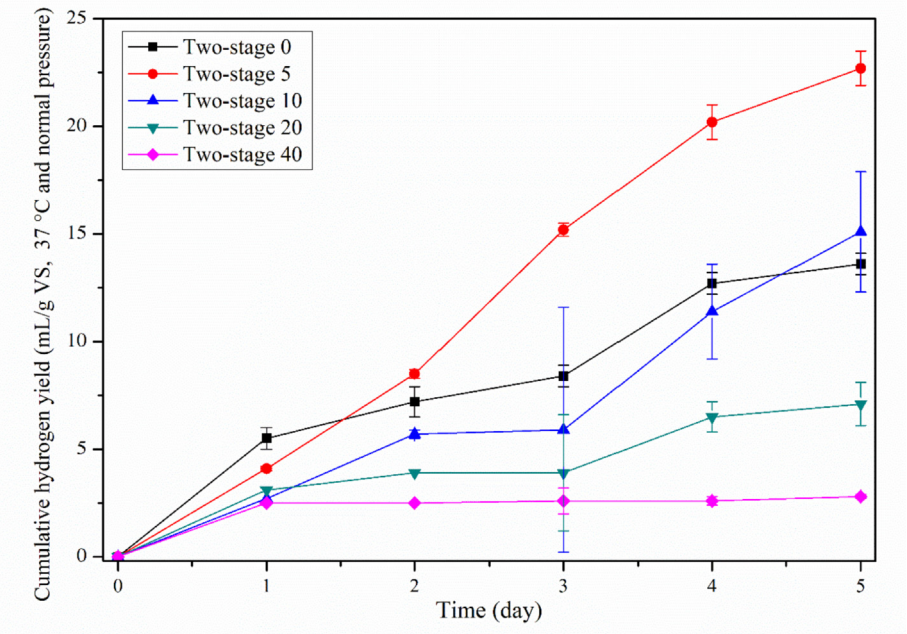


Fig. 2. Cumulative hydrogen yield vs. time during microaerobic hydrogen fermentation.

loading over 10 mL/g VS in this research).

3.2.2. VFA production

VFAs are the by-products of the hydrogen fermentation process. In current study, the VFA concentrations were measured daily using a GC (Gas Chromatography). The daily VFA concentrations are showed in Fig. 3. The VFA concentrations increased with the time. After 2 days of hydrogen fermentation, the VFA concentrations in AD reactors with oxygen addition were always higher than that in AD reactor without oxygen. The 5-day VFA concentrations with different levels of oxygen loading ranked in the order of Two-stage 5 > Two-stage 10 > Two-stage 20 > Two-stage 40 > Two-stage 0. The higher VFA concentrations in the presence of limited oxygen might be attributed to the higher hydrolysis efficiency under microaerobic condition (Li et al., 2016; Lim and Wang, 2013). However, excessive oxygen might result in decreased VFA concentration (e.g., Two-stage 20 and Two-stage 40), which could be explained by the further oxidization of VFAs under higher oxygen loadings (Nguyen et al., 2019).

During the whole hydrogen fermentation process, acetate and butyrate were the main VFAs, which took up more than 70% of the total VFAs. In addition, the proportion of acetate and butyrate in VFAs increased with the time. As reported, acetate and butyrate fermentation are the main metabolic pathways involved in hydrogen production (shown as Eqs. (4) and (5)) (Łukajtis et al., 2018). The main metabolic pathways of hydrogen production in the current study were still acetate and butyrate fermentation, which indicated that the microaerobic conditions did not alter the hydrogen production pathway.

$$C_6H_{12}O_6+2H_2O \rightarrow 2CH_3COOH+2CO_2+4H_2 \quad (5)$$

$$C_6H_{12}O_6 \rightarrow CH_3CH_2CH_2COOH+2CO_2+2H_2 \quad (6)$$

3.3. Methane production during the methane fermentation stage of the two-stage AD

3.3.1. Daily and cumulative methane yields

Due to the accumulation of VFAs during the hydrogen fermentation (Fig. 3), the pHs of the reactors dropped from 7.5 to 6.1–6.3. After re-inoculation of the fresh AD sludge for the methane fermentation in the 2nd stage, the pHs of the reactors were adjusted to 7.5 with NaOH (2 M) and HCl (2 M). Daily and cumulative methane yields during the methane fermentation are shown in Fig. 4. All the daily methane yields of two-stage AD and one-stage AD showed two peaks, and the daily methane yields after microaerobic hydrogen fermentation were generally higher than those of one-stage and Two-stage 0 at the initial 10 days of methane fermentation (Fig. 4a). These could benefit from the high VFA concentrations that were produced during the hydrogen fermentation (Fig. 3).

Methane fermentation stage lasted for 45 days. The cumulative methane yields of different AD scenarios showed significant difference ($p < 0.05$, Fig. 4b). The overall cumulative methane yields of two-stage AD were higher than that of one-stage. The respective cumulative methane yields of Two-stage 0, Two-stage 5, Two-stage 10, Two-stage 20 and Two-stage 40 were 264.5, 287.8, 296.0, 285.5, and 270.6 mL/g VS (37 °C and normal pressure), which were 3.6%, 12.7%, 15.9%, 11.8%, and 6.0% higher than that of one-stage, respectively. This result is consistent with previous reports (Akobi et al., 2016; Gioannis et al., 2017). When comparing the energy recovery from one- and two-stage anaerobic digestion of food waste, Gioannis et al. (2017) reported 19% higher methane yield was obtained from two-stage AD than that of one-stage AD. Similarly, Akobi et al. (2016) also found the two-stage AD of extruded lignocellulosic biomass obtained 11% higher methane yield than the one-stage.

Microaeration during the hydrogen fermentation stage generated positive effect on the methane yield. When the oxygen loadings during the hydrogen fermentation stage were 5, 10, 20, and 40 mL/g VS, the methane yields in the methane fermentation stage

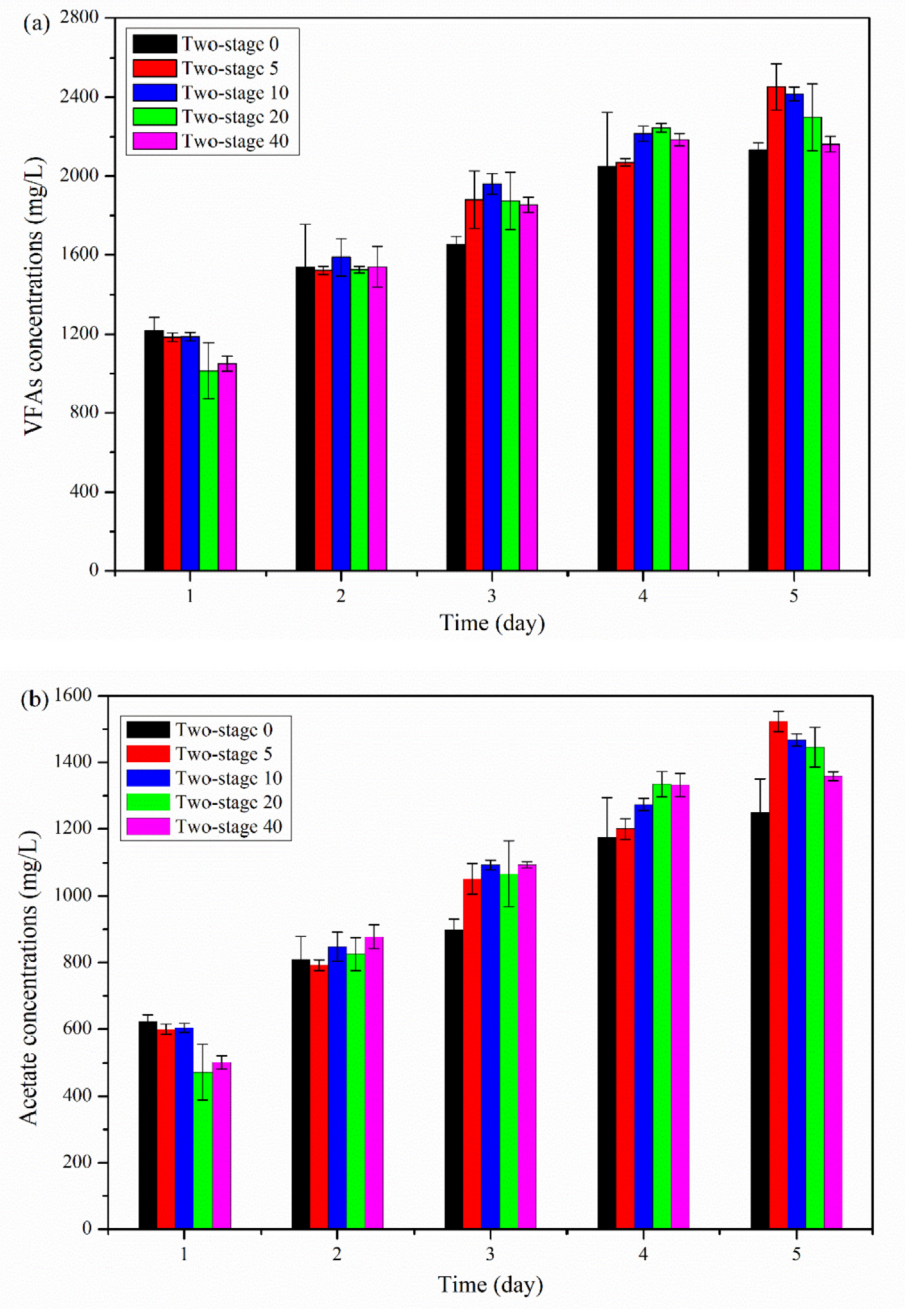


Fig. 3. Daily VFAs concentrations during hydrogen fermentation of two-stage AD: (a) total VFAs concentrations; (b) acetate concentrations.

were 287.8, 296.0, 285.5, and 270.6 mL/g VS (37 °C and normal pressure), which were 8.8%, 11.9%, 7.9%, and 2.3% higher than that of Two-stage 0, respectively. However, the positive effect really depended on the oxygen loading level. In this study, the oxygen loading of 10 mL/g VS during the hydrogen fermentation stage resulted in the maximum methane yield in the methane fermentation stage. After AD, the VS degradation rates of Two-stage 0, Two-stage 5, Two-stage 10, Two-stage 20, Two-stage 40 and One-stage were 65.3%, 73.3%, 74.2%, 71.2%, 69.2% and 60.2%, respectively. The VS degradation rate data was quite corresponding to the methane production results. Two-stage AD of corn straw obtained higher VS degradation rate, and microaeration during hydrogen fermentation could further improve the VS degradation rate. Maximum VS degradation rate was obtained when the oxygen loading during the hydrogen fermentation stage was 10 mL/g VS, which was 23.2% and 13.7% higher than those of One-stage and Two-stage without microaeration.

The positive effect of limited oxygen addition during the hydrogen fermentation on the methane yield could attribute to two reasons. Firstly, the microaerobic condition during the hydrogen fermentation generated more available substrate than the anaerobic condition (e.g., VFAs shown in Fig. 3) for methanogenesis. Secondly, the complex structure of the corn straw was destroyed more during microaerobic hydrogen fermentation, making the corn straw more available to biodegradation for methane production. However, high oxygen loading during the hydrogen fermentation stage did not necessarily lead to high methane yield, which could be explained by the excessive oxidation of the substrates. Similar

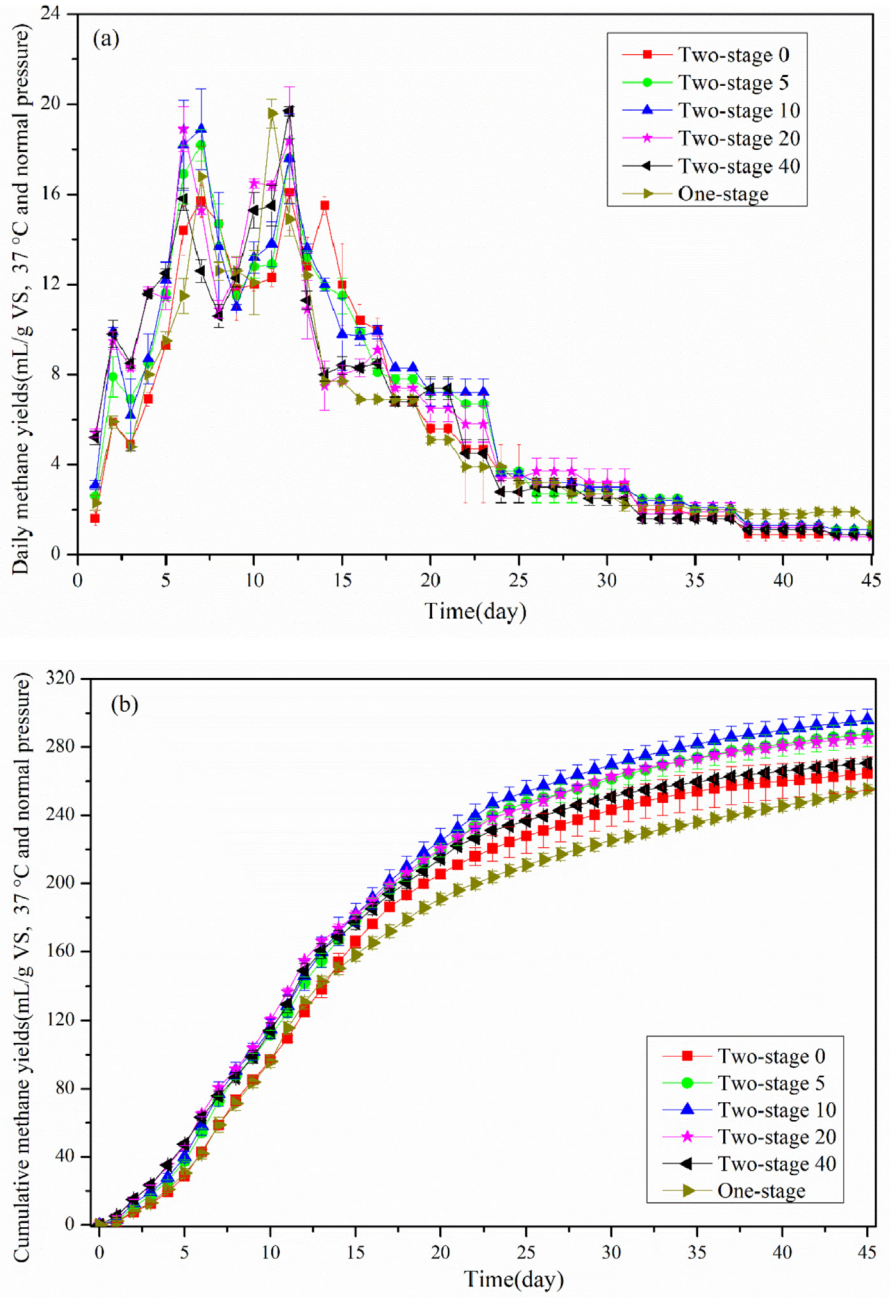


Fig. 4. Methane yield during methane fermentation: (a) daily methane yield and (b) cumulative methane yield.

results were obtained in both our previous studies (Fu et al., 2015, 2016b) and other research (Lim and Wang, 2013; Mshandete et al., 2005). According to the methane yields data in this study, a two-stage AD by combination of microaerobic hydrogen fermentation and anaerobic methane fermentation could be a potential strategy to produce bioenergy from corn straw, both the hydrogen and methane production efficiencies could be improved significantly. In addition, there is no additional chemical (except limited oxygen or air) needed during this procedure, which is more environmental friendly.

3.3.2. Methane production characteristics

The methane production characteristics were described with the parameters of *P*, *Rm*, λ, and T_{80} (T_{80} is technical time, which represents the time needed to produce 80% of the total methane yield). The *P*, *Rm* and λ were obtained by fitting the methane production data with the *Modified Gompertz equation* (Wang et al., 2018; Zhang et al., 2014). As shown in Table 2, all of the AD groups had a lag-phase. The span of the lag-phase of the two-stage AD was much lower than that of the one-stage AD probably because of more available substrates produced during the hydrogen fermentation stage of the two-stage AD (Yang et al., 2011). In addition, limited oxygen supply during hydrogen fermentation reduced the length of lag-phase of the two-stage AD. The maximum daily methane yields of the two-stage AD were also higher than that of the one-stage AD. The Two-stage 10 achieved the highest

Table 2
Methane production characteristics of the two-stage and one-stage AD.

Groups	P (mL/g VS, 37 °C and normal pressure)	Rm (mL/g VS per day, 37 °C and normal pressure)	λ (d)	R^2	T_{80} (d)
Two-stage 0	260.6 ± 1.1	13.9 ± 0.3	2.42 ± 0.23	0.998	21
Two-stage 5	283.0 ± 1.4	14.3 ± 0.4	2.32 ± 0.26	0.998	21
Two-stage 10	292.0 ± 1.4	14.4 ± 0.4	2.14 ± 0.26	0.998	21
Two-stage 20	281.8 ± 1.5	13.9 ± 0.4	1.55 ± 0.28	0.997	21
Two-stage 40	266.6 ± 1.1	13.8 ± 0.4	1.73 ± 0.24	0.998	20
One-stage	244.6 ± 1.9	12.6 ± 0.5	3.04 ± 0.41	0.995	24

methane production potential and daily methane yield, which reflects the benefit of microaeration during hydrogen fermentation to improve the methane production. The T_{80} of the two-stage AD was 3 or 4 days less than that of the one-stage AD, the results were quite consistent with the R_m data, which verified the higher methane production efficiencies of the two-stage AD.

3.4. Energy recovery from the two-stage and one-stage AD of corn straw

Generally, the AD process could be divide into three steps namely hydrolysis, acidification and methanogenesis, which involves four kinds of microorganisms, namely fermentative bacteria, hydrogen-producing bacteria, acetate-forming bacteria and methane-forming archaea (Luo et al., 2013; Xue et al., 2015). However, the optimal growth conditions of microorganisms involved in the AD process were different (e.g. fermentative bacteria, hydrogen-producing bacteria, acetate-forming bacteria prefer a slight acid condition; methane-forming archaea prefer neutral or slight alkali condition). Therefore, not all the microorganisms in AD are working at the optimum condition. Two-stage AD, in which the AD process is separated into the hydrolysis/acidogensis (stage I) and methanogenesis (stage II) that are optimized individually, could ensure the different microorganisms working at their optimum condition. In addition, the hydrogen fermentation stage of two-stage AD could also be considered as biological pretreatment method for organic solid waste, during the hydrogen fermentation hydrogen energy recovery as well as substrate pretreatment could be achieved simultaneously. Therefore, the two-stage AD performs better than one-stage AD in biogas production, which leads to higher energy recovery. Akobi et al. (2016) reported that the energy yields from the two-stage AD of extruded poplar wood were 33% (liquid streams obtained from poplar wood pretreated using the twin screw extrusion) and 18% (solid streams obtained from poplar wood pretreated using the twin screw extrusion) higher than those of the one-stage AD, respectively. Also in our previous study, the two-stage AD of vinasse resulted in 13% higher energy yield than that of the one-stage AD (Fu et al., 2017). In this study, energy recovery from the two-stage and one-stage AD of corn straw was calculated according to Eq. (2) and the resluts are shown in Table 3. Hydrogen yields of the Two-stage 0, Two-stage 5, Two-stage 10, Two-stage 20, and Two-stage 40 were 13.6, 22.7, 15.1, 7.1, and 2.8 mL/g VS (37 °C and normal pressure), respectively. Methane yields of the Two-stage 0, Two-stage 5, Two-stage 10, Two-stage 20, Two-stage 40, and One-stage were 264.5, 287.8, 296.0, 285.5, 270.6, and 255.3 mL/g VS (37 °C and normal pressure), respectively. The corresponding energy recoveries of the Two-stage 0, Two-stage 5, Two-stage 10, Two-stage 20, Two-stage 40, and the one-stage were 9.47, 10.41, 10.60, 10.13, 9.55, and 8.98 kJ/g VS. It is evident that the two-stage AD generated higher energy recovery, and the microaeration during the hydrogen fermentation further enhanced the energy recovery from the two-stage AD. The maximum energy recovery was obtained from the Two-stage 10, which was 12% and 18% higher than those of the Two-stage 0 and the one-stage, respectively.

4. Conclusions

Two-stage AD of corn straw including stepwise hydrogen fermentation and methane production improved the overall energy recovery statistically significantly compared to the one-stage AD. In addition, microaeration during the hydrogen fermentation further enhanced the hydrogen/methane yields and energy recovery. The beneficial effect of microaeration depended on the oxygen loading level. The maximum energy recovery was achieved by the combination of microaerobic hydrogen fermentation with the oxygen loading of 10 mL/g VS and methane fermentation, which was 18% higher than that of the one-stage AD. In addition, the length of the lag-phase and the total methane fermentation period were significantly reduced by using the two-stage AD and further reduced by adding microaeration. Therefore, the developed two-stage AD with stepwise microaerobic hydrogen fermentation and methanogenesis is a promising approach to maximize the energy recovery from AD of cellulose substrates. However, further researches also

Table 3
Energy recovery during the two-stage and one-stage AD of corn straw.

Groups	Energy recovery during hydrogen fermentation (kJ/g VS)	Energy recovery during methane fermentation (kJ/g VS)	Total energy recovery (kJ/g VS)	Energy recovery increase over the one-stage (%)
Two-stage 0	0.17 ± 0.01	9.30 ± 0.34	9.47 ± 0.35	5.46 ± 0.98
Two-stage 5	0.29 ± 0.04	10.12 ± 0.26	10.41 ± 0.30	15.92 ± 1.23
Two-stage 10	0.19 ± 0.02	10.41 ± 0.23	10.60 ± 0.25	18.04 ± 1.78
Two-stage 20	0.09 ± 0.02	10.04 ± 0.11	10.13 ± 0.13	12.81 ± 1.42
Two-stage 40	0.04 ± 0.01	9.51 ± 0.08	9.55 ± 0.09	6.35 ± 1.02
One-stage	0	8.98 ± 0.09	8.98 ± 0.09	0

need to conduct before its industry application. Firstly, the microaeration in this study was realized by injection of limited oxygen into the digester directly, it is not proper for industry application. Researches focus on how to realize microaeration in industry application are urgently needed. Secondly, the substrate used in this study was corn straw, the effects of microaeration on hydrogen/methane yields could be different based on the substrate. Therefore, effects of microaeration on hydrogen/methane yields during two-stage AD of different substrate also need to explore.

5. Author contributions section

Among the authors, Shan-Fei Fu and Hua Zou conceived the idea and designed the experiment. Yi Zheng and Zhen-Yu Wang guided the project. Shan-Fei Fu, Rong Liu and Wen-Xin Sun carried out the anaerobic digestion experiments, tested the experimental parameters and analyzed the experimental data. Shan-Fei Fu prepared the manuscript. Rong Zhu drew parts of the figures. Yi Zheng, Hua Zou and Zhen-Yu Wang revised the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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